820-0230 – Installation Instructions GC + PE Mixing Valves P/N 205-0121-02 (GE P/N 5156318)

People Required:
- 2 People

Procedure Timing:
- GC Mixing Valve Replacement: 3 hours
- PE Mixing Valve Replacement: 3 hours

Tools Required:
- 1-1/4 in. Open-end or Adjustable Wrench
- 7/16 in. Deep-well Socket or Wrench
- Wet/Dry Vacuum or Bucket for Draining
- Absorbent Towels

NOTE:
Old systems may have valve different from those in this kit. For that reason, hoses were supplied. Review entire procedure before beginning. Illustrations on pages 3 and 4 detail which valve/hose set should be replaced.

Removal Procedure:

1. Perform shutdown and apply Lockout/Tagout to the system in accordance with the GE *LOTO for Heat Exchanger Cabinet (HEC)* procedure (power needs to be removed for the electronics and facility water should be Locked and Tagged).
2. Open the door to HEC. If necessary, the door can be removed from its hinges by lifting up on the door.
3. Disconnect the 5-pin connector near the mixing valve body. The mixing valves are located at the top left side of the cabinet when viewing from the front. See Illustration 1 below.

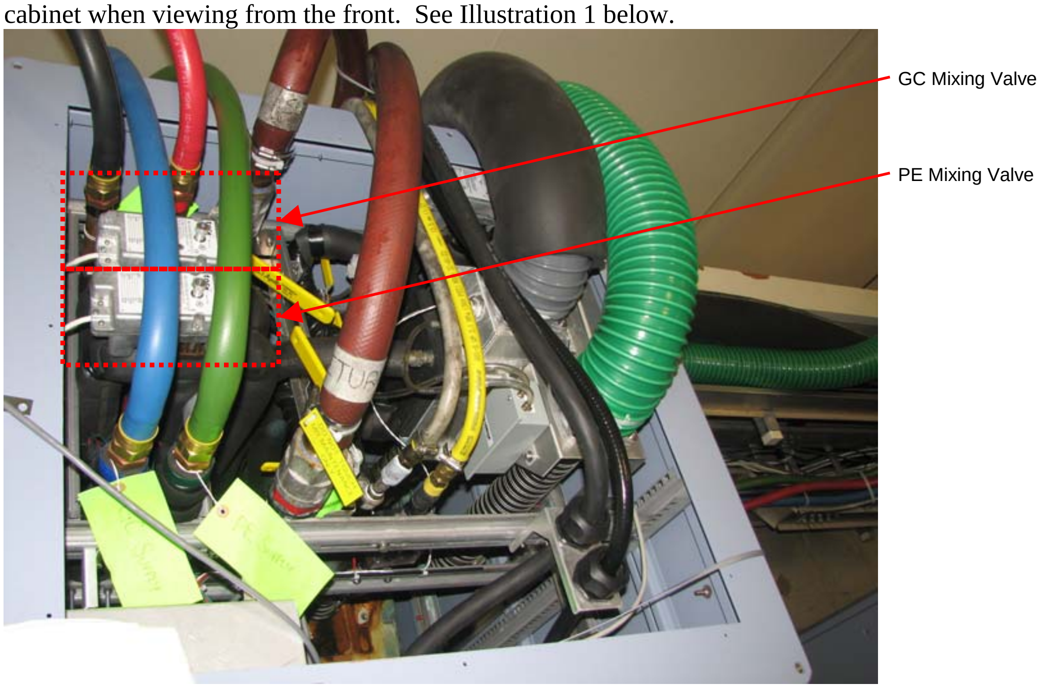


Illustration 1: Location of GC and PE Mixing Valves

4. Drain Facility side plumbing per GE draining procedures (*Coolant Draining*)

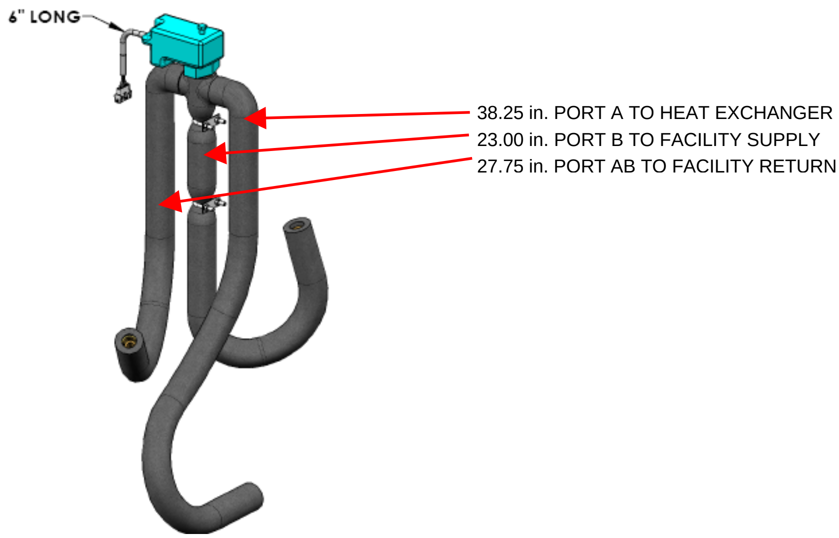


Illustration 2: Connections for Either GC or PE Three-Way Valve

NOTE: When removing items weighing more than 35 lb., the hoist system must be utilized.

5. Remove the Power Box and Chilled Air Blower from the front of the HEC. It may also be necessary to remove the Signal Box to gain access to connection points on Facility Return header may be required. (Refer to Lytron Procedures #820-0233 (Power Box), 820-0231 (Chilled Air Blower) and #820-0234 (Signal Box) for details on removing these parts.)

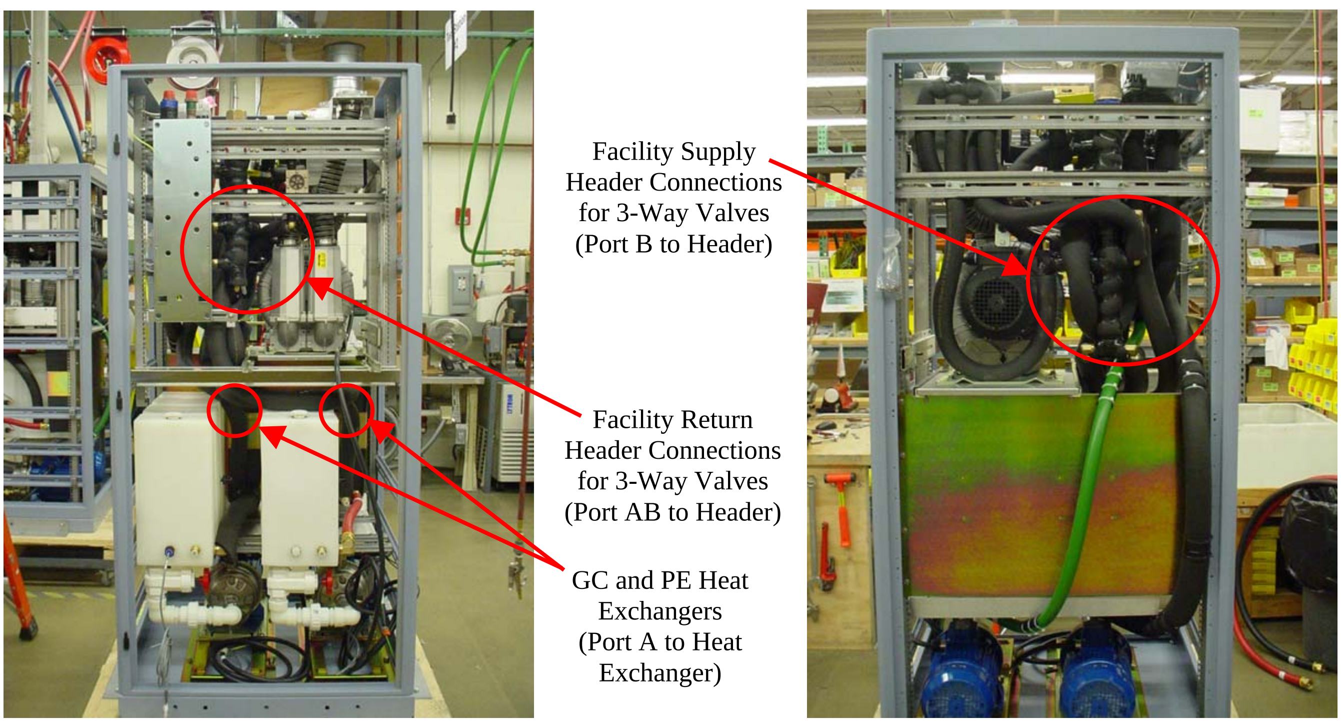


Illustration 3: Cabinet with Signal Box and Electrical Box Removed (Front and Rear Views)

6. Refer to the illustrations that follow for the locations to connect Ports A, AB, and B to the system.

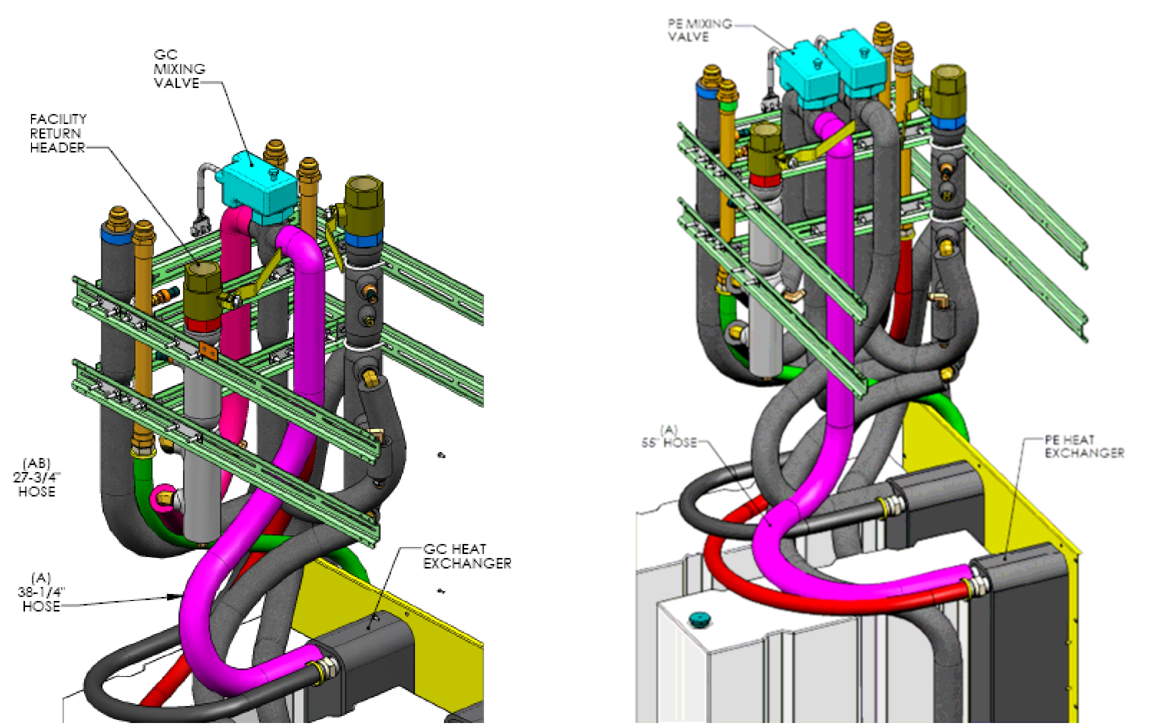


Illustration 4: Port A Installation to the Heat Exchangers

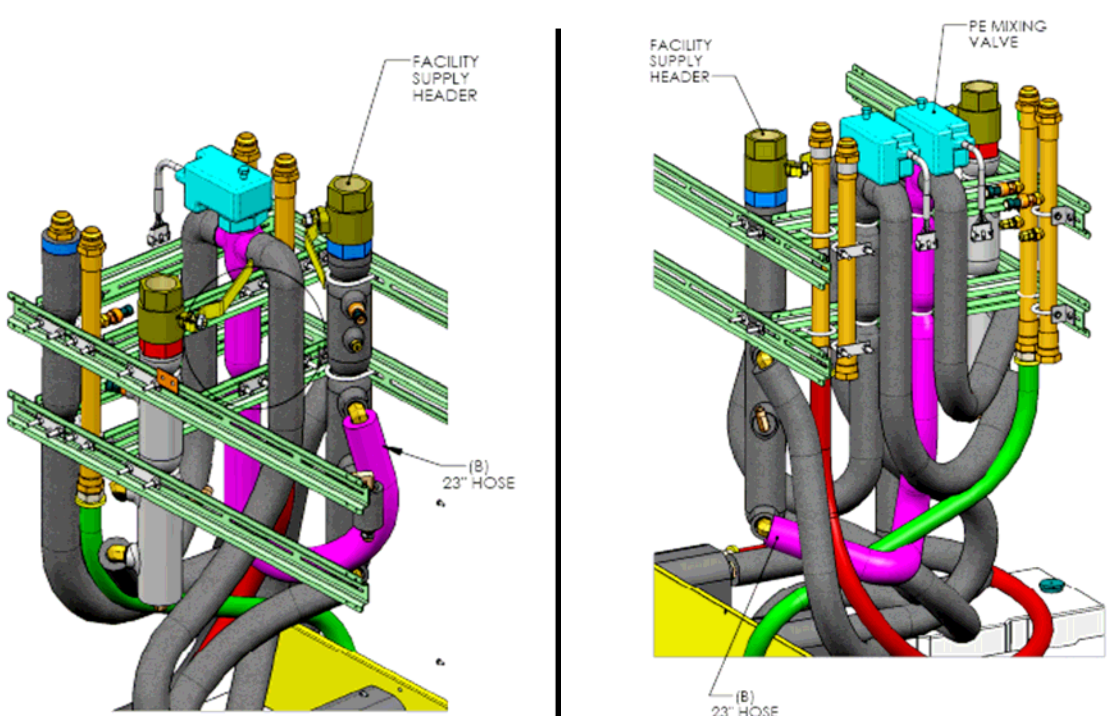
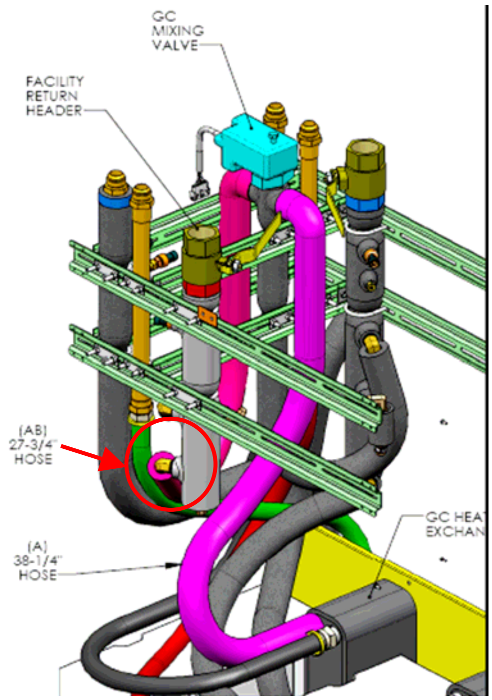


Illustration 5: Connecting Port B of GC and PE Three-Way Valves to Facility Supply Header

7. Using a 1-1/4 in. wrench, remove Port B from the Facility Supply. (It may be necessary to cut the insulation around the header.) There may be residual coolant in the hose that will need to be drained.
8. Using the 1-1/4 in. wrench, remove the Facility Return connection at the Header. (Insulation may need to be removed at the header and the hose drained.)
9. Use a 1-1/4 in. wrench to loosen the hose that attaches to the appropriate heat exchanger when looking from the front of the cabinet.
10. While supporting the valve assembly with one hand, remove the 1/4-20 captive nuts that secure the valve to the system using a U-bracket assembly.
11. Remove the valve assembly (valve and three attached hoses). Retain hardware for reinstallation.

Installation Procedure:

1. Attach the new valve assembly by securing it to the system frame with two captive nuts and the supportive U-bracket.
2. Reconnect the 5-pin connector to the mixing valve body.
3. Connect Hose A for either the GC or PE three-way valve to their respective Heat Exchangers using a 1-1/4 in. wrench. Tighten until hand tight, and then tighten an additional 1/4 turn. (Another wrench may be used behind the 1-1/4 in. wrench to protect the Heat Exchanger Flange.)
4. Attach Port B of the 3-way valves for either the GC or PE to the Facility Return header. Tighten until hand tight, and then tighten an additional 1/4 turn.
5. Attach hose AB for either the GC & PE Facility Supply Valves. Tighten until hand tight, and then tighten an additional 1/4 turn. (Another wrench may be needed behind the 1-1/4 in. wrench to protect the heat exchanger flange.)



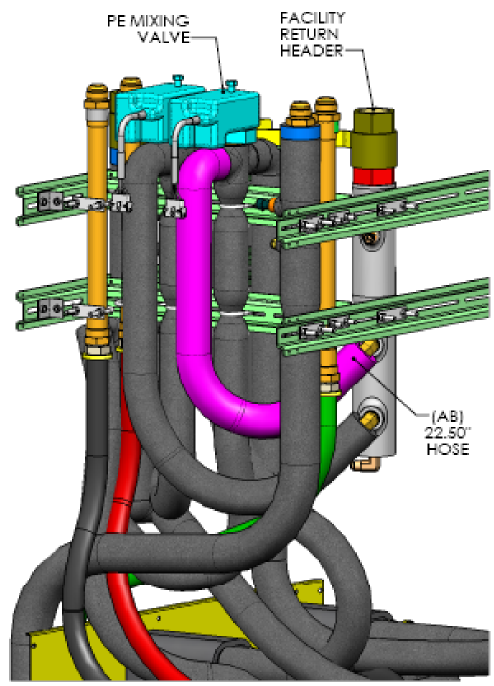


Illustration 6: Connection of AB Hose to GC and PE Heat Exchangers

6. Restart Facility water flow by removing LOTO from the Facility water ball valves and opening the valves fully.
7. Check the newly made connections for leaks.
8. Reinstall the Signal Box and Electrical Box.
9. Remove LOTO to the Electronics in the HEC.
10. Confirm the temperature set points and actual values are within 1 degree Celsius on the display on the front of the cabinet.